

Merge-Based Structured Pruning: a Geometric Theory with Certified and Expected-Error Guarantees

Working notes (theory companion to `studies/DESIGN_SPACE.md`)

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Abstract

We develop, from first principles, a theory of structured pruning by *merging* units of a ReLU layer. The theory proceeds in five steps: (1) a canonical, gauge-invariant parameterization of a ReLU unit as an oriented hyperplane with a loudness weight; (2) an associative merge operator on units that is exact on coincident units and whose pointwise error is localized to activation-disagreement regions; (3) a certified worst-case error bound whose accumulation is a valid stopping budget and whose squared form is the Ward clustering objective; (4) an average-case theory in the Hilbert space $L^2(\mu)$ of unit functions, in which merge damage, optimal fan-out repair, and the whole selection problem have closed forms under Gaussian input measures; (5) the identification of subset-plus-least-squares pruning (OBS/OSSCAR) as the medoid special case of the framework, which yields an adaptive method spanning both regimes. Every theorem is stated with its assumptions; a final section confronts each claim with the empirical record (E1–E13) and reports where assumptions failed and what that implied.

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1 Setting and notation

We consider one prunable layer of a feed-forward network. The layer receives inputs $x \in \mathbb{R}^d$, contains H ReLU units, and feeds a linear consumer.

Definition 1.1 (Layer, units, consumer). *Unit $i \in \{1, \dots, H\}$ has incoming weights $w_i \in \mathbb{R}^d$, bias $b_i \in \mathbb{R}$, and activation*

$$h_i(x) = \sigma(w_i^\top x + b_i), \quad \sigma(t) = \max\{0, t\}.$$

The consumer reads the layer through columns $c_i \in \mathbb{R}^m$ (row i of the next weight matrix, transposed), producing the pre-activation contribution

$$F(x) = \sum_{i=1}^H c_i h_i(x) \in \mathbb{R}^m.$$

Remark 1.2 (Convolutions). For a Conv $\rightarrow$ BN $\rightarrow$ ReLU block in evaluation mode, fold the Batch-Norm into effective weights $w^{\text{eff}} = (\gamma_{\text{bn}} / \sqrt{\sigma_{\text{run}}^2 + \epsilon}) w$ and $b^{\text{eff}} = \beta_{\text{bn}} - \gamma_{\text{bn}} \mu_{\text{run}} / \sqrt{\sigma_{\text{run}}^2 + \epsilon}$ (exact), let x range over im2col *patches*, and let c_i be the flattened kernel slice of the consumer conv (or the head column after global average pooling, which is linear per channel). Weight sharing means each *filter* is a single unit on patch space; every statement below applies verbatim.

Throughout, $x_0 \in \mathbb{R}^d$ is a reference point and $R_0 > 0$ a radius such that all admissible inputs satisfy $\|x - x_0\| \leq R_0$ (for images: the center and enclosing radius of the per-feature input box; for deeper layers: propagated bounds or calibration ranges). Only the *domain* (x_0, R_0) is assumed known in the worst-case theory; the average-case theory of Section 5 additionally uses an input *measure* μ .

2 Canonical parameterization and gauge

Definition 2.1 (Canonical coordinates). *For a unit with $\|w\| > 0$ define*

$$\alpha = \|w\| \quad (\text{gain}), \quad u = w / \|w\| \quad (\text{orientation}), \quad \rho = -b / \|w\| \quad (\text{signed offset}),$$

$$\gamma = u^\top x_0 - \rho \quad (\text{signed margin at } x_0), \quad a = \alpha \|c\| \quad (\text{loudness}).$$

The unit's pre-activation in normalized form is $s(x) = u^\top x - \rho = u^\top (x - x_0) + \gamma$, so that $h(x) = \alpha \sigma(s(x))$ and the activation boundary is the hyperplane $\{x : s(x) = 0\}$, at signed distance γ from x_0 along u .

Lemma 2.2 (Gauge invariance and completeness). *For every $t > 0$ the reparameterization $(w, b, c) \mapsto (tw, tb, c/t)$ leaves the contribution $ch(x)$ unchanged for all x . The invariants of this gauge action are exactly $(u, \rho, \alpha c)$; equivalently $(u, \gamma, \alpha c)$.*

Proof. By positive homogeneity, $\sigma(tz) = t \sigma(z)$ for $t > 0$, so $(c/t) \sigma(tw^\top x + tb) = c \sigma(w^\top x + b)$. Under the action, u and ρ are fixed and $\alpha c \mapsto (t\alpha)(c/t) = \alpha c$. Conversely $(u, \rho, \alpha c)$ determines $ch(x) = (\alpha c) \sigma(u^\top x - \rho)$, hence determines the orbit. $\square$

Lemma 2.3 (Sign-blindness of anchors). *The anchor $\beta := -(b / \|w\|) w / \|w\| = \rho u$ (the foot of the perpendicular from the origin to the boundary) is invariant under $(w, b) \mapsto (-w, -b)$, although this flip replaces the unit by its complementary unit (active on the opposite half-space). Moreover all boundaries through the origin have $\beta = 0$ regardless of orientation. Hence β alone does not determine the unit; the pair (u, γ) does.*

Proof. Under the flip, $\rho \mapsto -\rho$ and $u \mapsto -u$, so $\beta = \rho u$ is fixed, while the active half-space $\{s > 0\}$ is replaced by its complement. If $\rho = 0$ then $\beta = 0$ for every u . $\square$

Remark 2.4 (The covector cylinder). Encode a unit by $\tilde{q} = (R_0 u, \gamma) \in \mathbb{R}^{d+1}$. Since $\|R_0 u\| = R_0$, all units live on the cylinder $S^{d-1}(R_0) \times \mathbb{R}$: angular position is the boundary’s orientation (sign included — complementary units are antipodal), height is the boundary’s placement relative to the data. $\tilde{q}$ is the coefficient vector of the affine map $\delta \mapsto s(x_0 + R_0 \delta)$ on the unit ball, so Euclidean proximity of $\tilde{q}$ ’s controls uniform proximity of pre-activations (Theorem 4.1 makes this exact). This is the classical parameterization of oriented hyperplanes (in $d = 2$, Hough space).

3 The merge operator

Fix per-unit weights $a_i > 0$ (we will use the loudness $a_i = \alpha_i \|c_i\|$; the theory below holds for any positive weights).

Definition 3.1 (Cluster merge). *For a cluster $C \subseteq \{1, \dots, H\}$ define the covector sums*

$$S_u = \sum_{i \in C} a_i u_i, \quad S_\rho = \sum_{i \in C} a_i \rho_i, \quad S_c = \sum_{i \in C} \alpha_i c_i.$$

If $S_u \neq 0$, the merged unit is

$$\bar{u} = S_u / \|S_u\|, \quad \bar{\rho} = S_\rho / \|S_u\|, \quad \text{realized as } \bar{h}(x) = \sigma(\bar{u}^\top x - \bar{\rho})$$

with consumer column $\bar{c} = S_c$ (sum rule; refined in Section 6). The merged contribution is $\bar{F}_C(x) = S_c \sigma(\bar{s}(x))$, $\bar{s}(x) = \bar{u}^\top x - \bar{\rho}$.

Proposition 3.2 (Associativity). *Encode each unit by $\xi_i = a_i(u_i, \rho_i) \in \mathbb{R}^{d+1}$ and each merged unit by $\bar{\xi} = \|S_u\|(\bar{u}, \bar{\rho})$. Then $\bar{\xi} = \sum_{i \in C} \xi_i$, and likewise S_c is additive. Consequently merging is associative and order-independent: any sequence of pairwise merges whose final partition is $\mathcal{P}$ produces exactly the units obtained by merging each block of $\mathcal{P}$ at once.*

Proof. $\|S_u\| \bar{u} = S_u$ and $\|S_u\| \bar{\rho} = S_\rho$ by Definition 3.1, so $\bar{\xi} = (S_u, S_\rho) = \sum \xi_i$. Vector addition is associative and commutative; the realized parameters are a function of the sums only. $\square$

Proposition 3.3 (First-moment conservation). *The total covector $\Xi = \sum_k \xi_k$ (sum over current units, merged or not) and the total column sum $\sum_k \bar{c}_k$ are invariant under any sequence of merges. So is the “orientation energy” $\sum_k A_k R_0^2 \|\bar{u}_k\|^2 = R_0^2 \sum_i a_i$ (where A_k is the block’s weight sum), since every realized orientation is unit-norm.*

Proof. Immediate from Proposition 3.2 and additivity of the weights. $\square$

Theorem 3.4 (Exactness on coincident units, arbitrary fan-out). *If all units of C share one boundary and side, $(u_i, \rho_i) = (u, \rho)$ for all $i \in C$, then $\bar{F}_C(x) = F_C(x)$ for every $x \in \mathbb{R}^d$ and arbitrary columns c_i (any signs, any directions).*

Proof. $S_u = (\sum a_i) u$, hence $\bar{u} = u$, $\bar{\rho} = (\sum a_i \rho) / \sum a_i = \rho$. All units and the merged unit share the indicator $\mathbf{1}\{u^\top x > \rho\}$; on the active set, $F_C(x) = \sum_i c_i \alpha_i (u^\top x - \rho) = S_c (u^\top x - \rho) = \bar{F}_C(x)$, and both vanish elsewhere. $\square$

Theorem 3.5 (Pointwise error identity; scalar, slope-matched case). *Let $m = 1$, $a_i \geq 0$, $f_C(x) = \sum_{i \in C} a_i \sigma(s_i(x))$, and let the merged unit be slope-matched: $\bar{f}_C(x) = \sigma(\sum_i a_i s_i(x))$. Then for all x ,*

$$f_C(x) - \bar{f}_C(x) = \sum_{i \in C} a_i \sigma(s_i(x)) - \sigma\left(\sum_{i \in C} a_i s_i(x)\right) \geq 0,$$

with equality whenever all $s_i(x)$ share a sign. Consequently the error is supported on the activation-disagreement region $D_C = \{x : \exists i, j \in C, s_i(x) > 0 > s_j(x)\}$, a union of wedges between the cluster's boundaries.

Proof. Nonnegativity is subadditivity of σ on nonnegative combinations: $\sigma(\sum a_i s_i) \leq \sum a_i \sigma(s_i)$ by convexity and positive homogeneity. If all $s_i(x) \geq 0$ both sides equal $\sum a_i s_i(x)$; if all $s_i(x) \leq 0$ both sides vanish. $\square$

Corollary 3.6 (Saturation as the degenerate case). *If $\gamma_i \leq -R_0$ then $s_i(x) = u_i^\top(x - x_0) + \gamma_i \leq 0$ on the whole domain: the unit is identically zero there and its removal is exact. If $\gamma_i \geq R_0$ the unit is affine on the domain, $h_i(x) = \alpha_i(u_i^\top x - \rho_i)$, and can be absorbed exactly by any consumer able to represent affine functions of x (in practice: folded into downstream weights and biases). Dead-unit deletion, always-on folding, and redundancy merging are thus one mechanism: all are instances of Theorem 3.5 with empty or trivial disagreement region.*

Remark 3.7 (Sum rule vs. slope matching). The realized sum rule of Definition 3.1 gives, in the scalar case, $\bar{f} = A_C \sigma(\tilde{u}^\top x - \bar{\rho})$ with $\tilde{u} = S_u/A_C$, which *overshoots* the slope-matched unit by the factor $A_C/\|S_u\| \geq 1$ (equality iff all u_i coincide); the factor is $1 + O(\varepsilon_u^2)$ for angular dispersion ε_u . The projection update of Section 6 removes the overshoot optimally; the identity above is exact for the slope-matched realization.

4 Certified worst-case error

Theorem 4.1 (Certified merge bound, fan-out, centered). *Let C be merged as in Definition 3.1 with column $\bar{c} = S_c$. Then for every x with $\|x - x_0\| \leq R_0$,*

$$\|F_C(x) - \bar{F}_C(x)\| \leq \sum_{i \in C} \alpha_i \|c_i\| \left(R_0 \|u_i - \bar{u}\| + |\gamma_i - \bar{\gamma}| \right), \quad \bar{\gamma} := \bar{u}^\top x_0 - \bar{\rho}.$$

Proof. Since $\bar{c} = \sum_i \alpha_i c_i$,

$$F_C(x) - \bar{F}_C(x) = \sum_i \alpha_i c_i [\sigma(s_i(x)) - \sigma(\bar{s}(x))].$$

ReLU is 1-Lipschitz, so $|\sigma(s_i) - \sigma(\bar{s})| \leq |s_i - \bar{s}|$. Writing $x = x_0 + \delta$, $\|\delta\| \leq R_0$,

$$s_i(x) - \bar{s}(x) = (u_i - \bar{u})^\top \delta + (\gamma_i - \bar{\gamma}),$$

and Cauchy–Schwarz gives $|s_i - \bar{s}| \leq R_0 \|u_i - \bar{u}\| + |\gamma_i - \bar{\gamma}|$. The triangle inequality over i completes the proof. $\square$

Remark 4.2. The bound holds for *every* admissible input — it is a certificate, not an expectation — and it is computable from weights plus the domain (x_0, R_0) alone. Its accumulated value over all merged clusters is the stopping budget used by the pipeline. Centering at x_0 (rather than the origin) is essential in deep layers, where inputs are far from the origin.

Proposition 4.3 (Ward form of the bound). *For a pair $\{i, j\}$ merged at the weighted mean of the covectors $\tilde{q}_k = (R_0 u_k, \gamma_k)$ with weights a_k , the bound of Theorem 4.1 evaluates to $\frac{2a_i a_j}{a_i + a_j} d_1(i, j)$ with $d_1(i, j) = R_0 \|u_i - u_j\| \cdot \lambda + |\gamma_i - \gamma_j| \cdot (1 - \lambda)$ -type mean splits; more useful is the squared version: merging weighted points $\{(a_k, \tilde{q}_k)\}$ at their weighted mean reduces the weighted second moment $\sum_k a_k \|\tilde{q}_k\|^2$ by exactly the Ward cost*

$$\frac{a_i a_j}{a_i + a_j} \|\tilde{q}_i - \tilde{q}_j\|^2$$

(parallel-axis theorem). Greedy smallest-Ward-cost merging is therefore greedy minimization of a squared relative of the certified bound, and the accumulated second-moment loss of the covector cloud equals the accumulated Ward cost, independent of merge order.

Proof. Parallel axis: for weights m_k and points p_k with weighted mean $\bar{p}$ and $M = \sum m_k$, $\sum m_k \|p_k\|^2 - M \|\bar{p}\|^2 = \sum m_k \|p_k - \bar{p}\|^2$; for two points the right side is $\frac{m_1 m_2}{M} \|p_1 - p_2\|^2$ by direct expansion. Order independence follows from associativity of the weighted mean with additive weights. $\square$

Remark 4.4 (What the realized merge does to spectra). The realized merged covector is the sphere projection of the centroid ($\bar{u}$ is unit-norm), so: the first covector moment is conserved *exactly* (Proposition 3.3); the orientation block of the weighted second moment is conserved *mechanically*; and the parallel-axis identity for the remaining part holds to second order in cluster dispersion. Hence “spectral stability of the layer” is meaningful only for the *loudness-weighted, centered* second moment; unweighted or anchor-based Grams change even under exact merges (they lose the multiplicity a_i) and are dominated by large- $|\gamma|$ (saturated, functionally irrelevant) units.

Proposition 4.5 (Composition across layers). *Let the network be $f = g_\ell \circ f_\ell$ where f_ℓ contains the pruned layer and g_ℓ (the downstream network) is L_ℓ -Lipschitz on a set containing the reachable sets of both the original and pruned f_ℓ . If each pruned layer ℓ incurs error at most ε_ℓ in the consumer’s pre-activation (uniformly on its admissible inputs), then*

$$\|\hat{f}(x) - f(x)\| \leq \sum_\ell L_\ell \varepsilon_\ell \quad \text{for all admissible } x,$$

where $\hat{f}$ prunes all layers and L_ℓ is the Lipschitz constant of everything downstream of layer ℓ ’s consumer.

Proof. Telescope: replace one layer at a time, $\hat{f} - f = \sum_\ell (f^{(\leq \ell)} - f^{(\leq \ell-1)})$ where $f^{(\leq \ell)}$ prunes layers 1.. ℓ . Each summand differs only in layer ℓ ; its norm is at most $L_\ell \varepsilon_\ell$ by the Lipschitz property, provided ε_ℓ holds on the (possibly shifted) inputs reaching layer ℓ — guaranteed when the domain radius used for ε_ℓ covers the pruned network’s reachable set, e.g. by enlarging R_0 by the accumulated upstream error. $\square$

5 Average-case theory: the functional Hilbert space

Worst-case bounds defend every point of the domain equally; an *average-case* theory weights inputs by a probability measure μ on $\mathbb{R}^d$ with finite second moments.

Definition 5.1 (Unit functions and kernel). *Let $h_i(x) = \alpha_i \sigma(u_i^\top x - \rho_i) \in L^2(\mu)$ and define the kernel $K_{ij} = \langle h_i, h_j \rangle_{L^2(\mu)} = \mathbb{E}_\mu[h_i(x)h_j(x)]$. For layer contributions, the relevant Gram is $\tilde{K}_{ij} = (c_i^\top c_j) K_{ij}$, since $\mathbb{E} \|\sum_i c_i h_i\|^2 = \sum_{ij} \tilde{K}_{ij}$.*

Proposition 5.2 (Exact expected merge damage). *Replacing $\{(c_i, h_i)\}_{i \in C}$ by a single unit $(\bar{c}, \bar{h})$ incurs*

$$\mathbb{E}_\mu \left\| \sum_{i \in C} c_i h_i - \bar{c} \bar{h} \right\|^2 = \sum_{i, j \in C} (c_i^\top c_j) K_{ij} - 2 \sum_{i \in C} (c_i^\top \bar{c}) \langle h_i, \bar{h} \rangle + \|\bar{c}\|^2 \langle \bar{h}, \bar{h} \rangle,$$

a closed quadratic form in kernel evaluations. In particular the exact expected damage of every candidate pair merge — and of any surgery — is computable without forward passes once K is available.

Proof. Expand the square and exchange expectation with the finite sums. $\square$

5.1 Closed forms under Gaussian measures

Let $\mu = \mathcal{N}(\mu_0, \Sigma)$. Each pre-activation $t_i = u_i^\top x - \rho_i$ is Gaussian with mean $m_i = u_i^\top \mu_0 - \rho_i$ and covariance $\text{Cov}(t_i, t_j) = u_i^\top \Sigma u_j$; hence every kernel entry reduces to the bivariate quantity $G(a, b, c) = \mathbb{E}[(X - a)_+(Y - b)_+]$ for standard bivariate normal (X, Y) with correlation c , via $K_{ij} = \alpha_i \alpha_j s_i s_j G(-m_i/s_i, -m_j/s_j, \text{corr}(t_i, t_j))$, $s_i^2 = u_i^\top \Sigma u_i$.

Theorem 5.3 (Closed form of G). *Let $s = \sqrt{1 - c^2} > 0$, φ, Φ the standard normal density and cdf, $\bar{\Phi} = 1 - \Phi$, $\beta_a = (b - ca)/s$, $\alpha_b = (a - cb)/s$, and $L(a, b; c) = \Pr(X > a, Y > b)$. Then*

$$G(a, b, c) = (c + ab) L(a, b; c) - b \varphi(a) \bar{\Phi}(\beta_a) - a \varphi(b) \bar{\Phi}(\alpha_b) + s \varphi(b) \varphi(\alpha_b).$$

Special cases: $G(a, a, 1) = (1 + a^2) \bar{\Phi}(a) - a \varphi(a)$; $G(0, 0, \cos \theta) = \frac{1}{2\pi} (\sin \theta + (\pi - \theta) \cos \theta)$, the arc-cosine kernel of Cho–Saul.

Proof. Write $Y = cX + sZ$ with $Z \perp X$ standard normal, and let $\mathbf{1} = \mathbf{1}\{X > a, Y > b\}$. Expanding $G = \mathbb{E}[XY\mathbf{1}] - b\mathbb{E}[X\mathbf{1}] - a\mathbb{E}[Y\mathbf{1}] + abL$, it suffices to compute the truncated moments. The key identity

$$\varphi(x) \varphi\left(\frac{b - cx}{s}\right) = \varphi(b) \varphi\left(\frac{x - cb}{s}\right)$$

follows by completing the square in the exponent. Using it: (i) $\mathbb{E}[X\mathbf{1}] = \varphi(a) \bar{\Phi}(\beta_a) + c \varphi(b) \bar{\Phi}(\alpha_b)$, by integration by parts on $\int_a^\infty x \varphi(x) \bar{\Phi}(\frac{b - cx}{s}) dx$; (ii) $\mathbb{E}[Y\mathbf{1}] = \varphi(b) \bar{\Phi}(\alpha_b) + c \varphi(a) \bar{\Phi}(\beta_a)$ by symmetry; (iii) $\mathbb{E}[XZ\mathbf{1}] = s \varphi(b) (cb \bar{\Phi}(\alpha_b) + s \varphi(\alpha_b))$, obtained by conditioning on X ($\mathbb{E}[Z\mathbf{1}\{Z > g(X)\} | X] = \varphi(g(X))$ with $g(x) = (b - cx)/s$), substituting the key identity, and evaluating $\int_a^\infty x \varphi(\frac{x - cb}{s}) dx = s(cb \bar{\Phi}(\alpha_b) + s \varphi(\alpha_b))$; (iv) $\mathbb{E}[X^2\mathbf{1}] = L + a \varphi(a) \bar{\Phi}(\beta_a) + c \varphi(b) (cb \bar{\Phi}(\alpha_b) + s \varphi(\alpha_b))$, by parts using $\frac{d}{dx}(-\varphi) = x\varphi$. Then $\mathbb{E}[XY\mathbf{1}] = c\mathbb{E}[X^2\mathbf{1}] + s\mathbb{E}[XZ\mathbf{1}] = cL + ca \varphi(a) \bar{\Phi}(\beta_a) + cb \varphi(b) \bar{\Phi}(\alpha_b) + s \varphi(b) \varphi(\alpha_b)$, using $c^2 + s^2 = 1$. Assembling the four moments yields the stated formula. For $c = 1$, $Y = X$ and $G(a, a, 1) = \mathbb{E}[(X - a)_+^2]$ evaluates directly; for $a = b = 0$, $L(0, 0; c) = \frac{1}{4} + \frac{\arcsin c}{2\pi}$ gives the arc-cosine form. $\square$

Remark 5.4 (Which Gaussian?). Three instantiations, in decreasing ignorance: (i) *isotropic*: $\Sigma = \frac{R_0^2}{d+2} I$, matching the second moments of the uniform ball (for z uniform on B_{R_0} , $\mathbb{E}[(u^\top(z - x_0))^2] = R_0^2/(d+2)$); (ii) *product/box*: $\Sigma = \text{diag}(r^2/3)$ from per-feature ranges; (iii) *matched*: (μ_0, Σ) estimated from unlabeled calibration inputs. The choice is the entire empirical risk of the average-case theory: the formulas are exact *under* μ , and mis-specified μ produces confidently wrong scores (Section 9, E2/E4/E10). Trained units aim at correlated input directions, so any product measure underestimates the variance of t_i ; full covariance is the minimum sufficient correction, and for strongly non-Gaussian inputs (conv patches: a point mass of background plus sparse rectified foreground) even matched Gaussians fail and the empirical measure must be used.

6 Optimal fan-out repair

Merging fixes the surviving *units*; the consumer's columns remain free. All repairs below are orthogonal projections in $L^2(\mu)$ and therefore optimal in expected squared error under μ .

Proposition 6.1 (Per-cluster projection). *Given the merged unit $\bar{h}$, the column minimizing $\mathbb{E} \left\| \sum_{i \in C} c_i h_i - \bar{c} \bar{h} \right\|^2$ is*

$$\bar{c}^\star = \frac{\sum_{i \in C} c_i \langle h_i, \bar{h} \rangle}{\langle \bar{h}, \bar{h} \rangle}.$$

For $m = 1$ and the choice $a_i = \alpha_i c_i > 0$ this reduces to the slope-matched rule of Theorem 3.5 up to $O(\varepsilon^2)$ (and exactly in the co-active region).

Proof. The objective is a quadratic in $\bar{c}$ minimized coordinate-wise by projecting each output coordinate onto $\text{span}\{\bar{h}\}$. $\square$

Theorem 6.2 (Global least-squares repair). *Let $\{\hat{h}_k\}_{k=1}^K$ be the surviving (merged or original) units and $G_{k\ell}^{kk} = \langle \hat{h}_k, \hat{h}_\ell \rangle$, $G_{ki}^{ko} = \langle \hat{h}_k, h_i \rangle$. The columns minimizing $\mathbb{E} \left\| \sum_i c_i h_i - \sum_k \hat{c}_k \hat{h}_k \right\|^2$ solve the normal equations*

$$G^{kk} \hat{C} = G^{ko} C,$$

with $\hat{C}, C$ the matrices of columns. Under Gaussian μ every entry of G^{kk}, G^{ko} is a closed-form evaluation of Theorem 5.3; under the empirical measure of N samples the same equations are ridge least squares on activations (the repair used by OBS-family methods).

Proof. Orthogonal projection of the vector-valued function $\sum_i c_i h_i$ onto $\text{span}\{\hat{h}_1, \dots, \hat{h}_K\}$, coordinate-wise in the output index. $\square$

Corollary 6.3 (Bias compensation). *If the consumer has a bias (or the next normalization layer can absorb a constant), replacing it by $b + \mathbb{E}_\mu[\text{residual}]$ further reduces the expected squared error by $\|\mathbb{E}_\mu[\text{residual}]\|^2$, by the mean/centered orthogonal decomposition. For the slope-matched merge, the residual is pointwise nonnegative (Theorem 3.5), so this correction is systematically nonzero. Under Gaussian μ , $\mathbb{E}[\sigma(t)] = s \varphi(m/s) + m \Phi(m/s)$ gives it in closed form.*

Remark 6.4 (Parametric vs. empirical Grams). The two instantiations of Theorem 6.2 differ only in the estimator of the Gram matrices: analytic under a fitted μ ($O(d^2)$ parameters, well-conditioned, exactly correct iff μ is), versus empirical from N samples (unbiased for the true measure, rank $\leq N$, variance-limited). When second moments are sufficient statistics of the true input distribution the two coincide as N grows and the parametric form regularizes at small N ; when the input law is far from Gaussian the parametric form is inconsistent and the empirical form must be preferred. This single dichotomy explains the architecture split in the empirical record (E9 vs. E10).

7 Relation to OBS-family pruning (OSSCAR)

Subset-selection methods choose a support $S \subseteq \{1..H\}$, zero the complement, and re-solve the consumer on S : with empirical Gram $H = X^\top X$ (consumer inputs X) and target $G = HB$, they minimize the quadratic $Q(W) = \frac{1}{2} \text{Tr}(W^\top H W) - \text{Tr}(G^\top W)$ over W supported on S 's rows.

Proposition 7.1 (OBS removal score). *Let $W^\star = H^{-1}G$ be the unconstrained minimizer and let unit j own one row (the group case is analogous with block inverses). The minimal objective increase from constraining $W_j = 0$, with all other rows re-optimized, is*

$$\Delta Q_j = \frac{1}{2} \frac{\|W_j^\star\|^2}{[H^{-1}]_{jj}}.$$

Proof. Minimize $Q(W^\star + \Delta)$ over Δ subject to $e_j^\top (W^\star + \Delta) = 0$. Since $\nabla Q(W^\star) = 0$, the increase is $\frac{1}{2} \text{Tr}(\Delta^\top H \Delta)$; the Lagrangian stationarity gives $\Delta = -H^{-1}e_j\lambda^\top$ with $\lambda = W^{\star\top}e_j/[H^{-1}]_{jj}$, and substituting yields the stated value. $\square$

Remark 7.2 (Placement inside the framework). (i) OSSCAR’s repair is Theorem 6.2 under the empirical measure. (ii) Its selection score is *conditional* (cost of removal given optimal compensation), whereas the merge costs of Sections 4–5 are local; its swap stage partially restores global optimality. (iii) Its dictionary is the *subset* family: exactly the special case of Definition 3.1 in which each cluster is realized by a member (medoid) rather than the mean. Consequently: where genuine clusters exist, mean realization strictly enlarges the dictionary (and empirically wins, E8); where they do not, mean units interpolate between dissimilar detectors and the medoid/subset realization is correct (E10–E13). The adaptive rule — mean when the cluster’s certificate (Theorem 4.1) is small, medoid otherwise — contains both.

8 The operating pipeline

Selection. Build the merge dendrogram greedily: repeatedly merge the pair (cluster) of minimal cost, where cost is the Ward form (Proposition 4.3; domain knowledge only) or the exact expected damage (Proposition 5.2 under the best available μ). By Proposition 3.2 the trajectory is a dendrogram: every width is a prefix, computed once.

Realization. Per cluster: mean unit (Definition 3.1) when its certified dispersion is small, medoid otherwise. Fan-out: global repair (Theorem 6.2) with the Gram estimator matched to the input law (parametric for MLP-type inputs; empirical for conv patches), plus bias compensation (Corollary 6.3); optionally one swap refinement pass over the support (Section 7).

Stopping. Accumulate the certificate of Theorem 4.1 (and/or the expected damage) and stop at a budget; per-layer budgets compose by Proposition 4.5. Thresholding pairwise similarity is strictly dominated: it cannot control accumulation and its scale is layer-dependent.

Tiers. The same pipeline degrades gracefully with information: domain only $\rightarrow$ Ward costs + certificate (fully data-free); $+O(d^2)$ moments from $\approx 10^2$ unlabeled inputs $\rightarrow$ closed-form expected damage and repair; $+ \text{raw calibration activations} \rightarrow$ empirical Grams and swap refinement (OBS-equivalent repair).

9 Confrontation with the empirical record

Each theoretical claim above was tested; the ledger (E1–E13, in `studies/DESIGN_SPACE.md`) is summarized against the theory. Numbers are medians over seeds; “capacity” is the fraction of units removable within a stated accuracy budget.

Theory	Evidence
Certificate (Thm 4.1) tracks true damage	E1: Spearman 0.99–1.0 between accumulated bound and measured layer error, all datasets. E6: as an <i>absolute</i> threshold it transfers across seeds/layers within a dataset (0.6–0.77 of oracle capacity at zero violations) but its pessimism factor is domain-dependent — ordering is certified, calibration is per-domain.
Conservation laws (Prop 3.3)	E1 and Figure 3: first moment conserved to 2×10^{-16} along every trajectory; orientation-block energy conserved mechanically (this <i>predicted</i> the blindness of trace-based spectral stopping rules, confirmed).
Exactness and identity (Thms 3.4, 3.5)	E3: machine-precision on coincident units with arbitrary fan-out; associativity verified bit-wise.
Ward = squared certificate (Prop 4.3)	E1/E7: greedy Ward selection is the robust domain-only policy; its costs are <i>not</i> comparable across layers (worst-case units), so cross-layer allocation must use expected damage or equal fractions (E7).
Measure choice is the whole risk (Rem 5.4)	E2: isotropic μ collapses on MNIST layer 0 (real projection spread $2.3\times$ the model). E4: product measures also fail; matched covariance repairs it and dominates (27–2–1 vs. Ward on layer error). E10: on conv patches even matched Gaussians fail (kernel repair degrades $2\text{--}3\times$ faster than empirical at identical structure).
Moment sufficiency (Rem 6.4)	E9: on MLPs, parametric and empirical Grams give <i>identical</i> capacities at $N=128$ in every cell — second moments capture everything the samples know. E10: the same comparison fails on patches — non-Gaussianity, as predicted by the dichotomy.
Global repair (Thm 6.2)	E8: +15–22 points of joint capacity over the sum rule on MLPs at equal data; exceeds the allocation <i>oracle</i> under the old surgery — repair dominates allocation.
Dictionary placement (Rem 7.2)	E8: mean realization beats medoid/survivor by +7–12 points where clusters exist (MLPs). E11: removal sets of our selection and OS-SCAR overlap far above chance (up to 0.94 vs. 0.50). E12/E13: medoid + empirical LS + swap on our supports reaches <i>exact</i> OS-SCAR parity on CNNs (gap decomposition: surgery +0.31, greedy myopia +0.07, remainder 0).
Saturation corollary (Cor 3.6)	Beta-saturation study: data-free certificates for dead/always-on units; BatchNorm suppresses saturation (zero certified-saturated filters in conv nets), so the merging channel carries conv pruning.
Composition (Prop 4.5)	E7: joint pruning of all layers loses almost nothing relative to per-layer capacities (64–70% of all hidden units at -1pt on MLPs); intact-model dendrograms suffice (sequential recomputation adds nothing).

9.1 Key results in figures and tables

Figures 1–2 and Tables 1–4 display the primary results; raw per-step trajectories live in the repository outputs directory. Two additional numbers complete the record: on pretrained ResNet-18 (Imagenette, E5), per-block filter counts removable at -1pt were 35/35/71/71/127/143/287/95 of 64/64/128/128/256/256/512/512 for Ward and 47/19/63/47/95/127/255/287 for the matched measure (task over-provisioning, not weight-space redundancy: at 5% layer error every arm manages only 1–3% of filters); and stopping-rule transfer (E6) gave 0.6–0.77 of oracle capacity at zero violations within a domain, with only the normalized cumulative expected damage transferring safely across domains (≈ 0.5 capture at 0–3% violations).

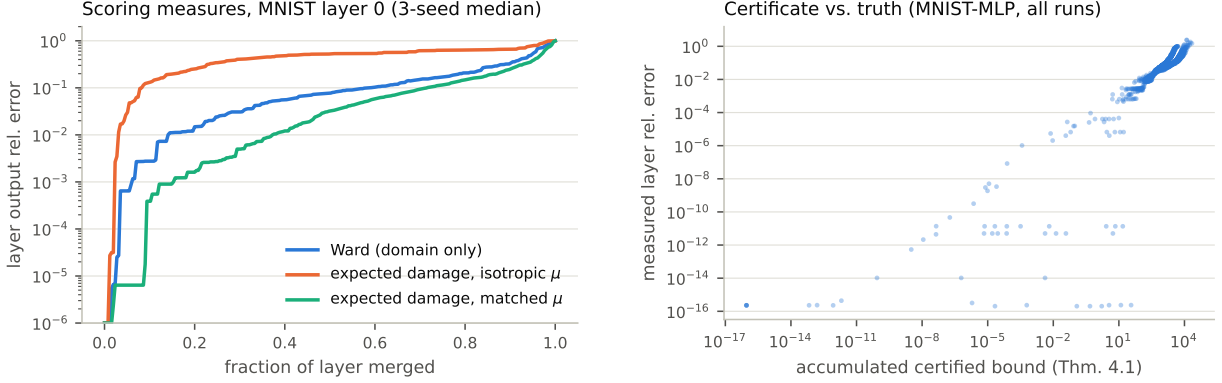


Figure 1: **Left (E2/E4)**: the measure is the whole risk of the average-case theory. On MNIST layer 0 the isotropic measure merges “confidently wrong” (error explodes almost immediately); the matched measure dominates the domain-only Ward metric; all three share the identical merge surgery. **Right (E1)**: the certificate of Theorem 4.1 against measured layer error across all layers, seeds, and merge steps (log-log): the relationship is monotone over ~ 15 orders of magnitude (Spearman 0.99–1.0 per trajectory). The horizontal band at 10^{-16} is the machine-precision floor of exact merges (Theorem 3.4): the bound is pessimistic there, never optimistic.

Table 1: **E4 (Phase A)**: scoring measures on MNIST-MLP layer 0 (medians over 3 seeds). Capacity = fraction of the layer mergeable within the stated budget. Right: wins–ties–losses vs. Ward on layer-local capacity (5% budget) over all 30 (dataset, seed, layer) runs.

scoring measure	cap.	@1% err	@5% err	@-1pt acc	W–T–L vs Ward
Ward, sphere (1A1)		0.141	0.366	0.616	—
Ward, ellipsoid (1A2)		0.137	0.325	0.706	6–5–19
expected damage, isotropic (1B1)		0.031	0.063	0.146	10–1–19
expected damage, box-CLT (1B3)		0.031	0.055	0.146	7–0–23
expected damage, matched (1B4)		0.376	0.576	0.769	27–2–1

Honest failures and their theoretical meaning. Three assumptions failed measurably, each in the direction the theory anticipates once the assumption is stated: (a) isotropy and independence of input coordinates (Remark 5.4) — falsified wherever inputs are structured; (b) Gaussianity of conv patch distributions — falsified, forcing the empirical estimator on conv models; (c) the existence of cluster structure in well-trained conv filters — absent under BatchNorm at scale (pretrained ResNet-18: 1–3% of filters mergeable at 5% layer error), so that on such models pruning capacity is *task* over-provisioning rather than weight-space redundancy, and the medoid/subset realization with data-based repair is the correct operating point. None of these failures touches the theorems; they locate the boundaries of their hypotheses.

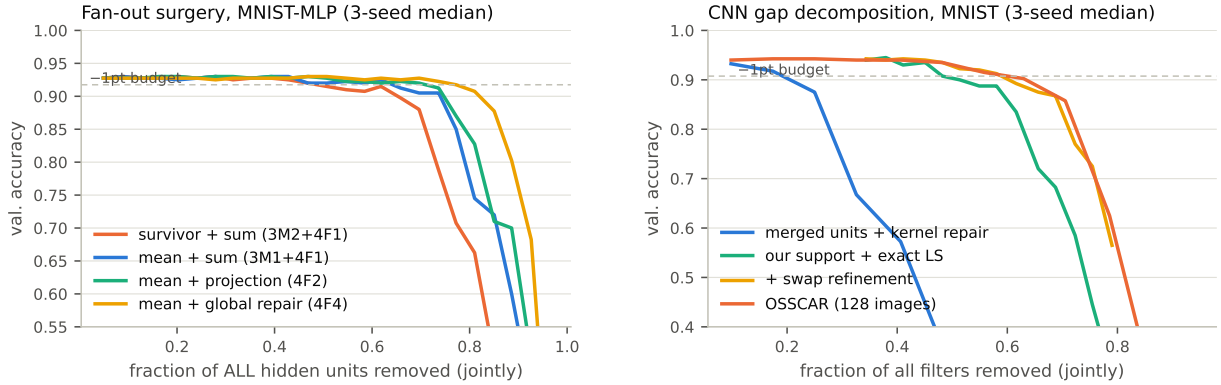


Figure 2: **Left (E8)**: the fan-out surgery hierarchy on MLPs (partitions fixed across variants): survivor < mean < projection < global repair (Theorem 6.2), the last worth +15 accuracy-capacity points. **Right (E10–E13)**: the CNN gap decomposition. The merged-unit/kernel pipeline collapses early; keeping our supports but switching to the exact-LS repair recovers most of the gap; one swap refinement pass reaches OSSCAR exactly — selection was never the problem (cf. the overlap of Table 4).

Table 2: **E8 (Phase B)**: fan-out surgery, joint equal-fraction pruning of all layers, func_matched partitions fixed across variants. Capacity = fraction of *all* hidden units removed within the budget (3-seed medians). The global repair exceeds even the cross-layer allocation grid *oracle* under the sum surgery (0.700 on MNIST, E7).

surgery	MNIST		Fashion	
	@−0.5pt	@−1pt	@−0.5pt	@−1pt
survivor + sum (3M2+4F1)	0.429	0.547	0.429	0.429
mean + sum (3M1+4F1)	0.585	0.618	0.469	0.547
mean + projection (4F2)	0.658	0.696	0.585	0.585
mean + global repair (4F4)	0.737	0.772	0.772	0.772

Table 3: **E9 (Phase D, MLPs)**: ours vs. OSSCAR at matched data budgets, joint pruning, capacity within budget (3-seed medians; width grid resolution ± 0.08). *hybrid128* = our structure with empirical Grams: identical to the parametric kernel in every cell — second moments are sufficient statistics here (Remark 6.4).

arm	MNIST			Fashion		
	−0.5pt	−1pt	−2pt	−0.5pt	−1pt	−2pt
ours (128 imgs, moments)	0.712	0.712	0.790	0.790	0.790	0.790
hybrid128 (empirical)	0.712	0.712	0.790	0.790	0.790	0.790
OSSCAR (128 imgs)	0.712	0.790	0.790	0.712	0.790	0.790
OSSCAR (full 2000)	0.790	0.790	0.868	0.712	0.790	0.790

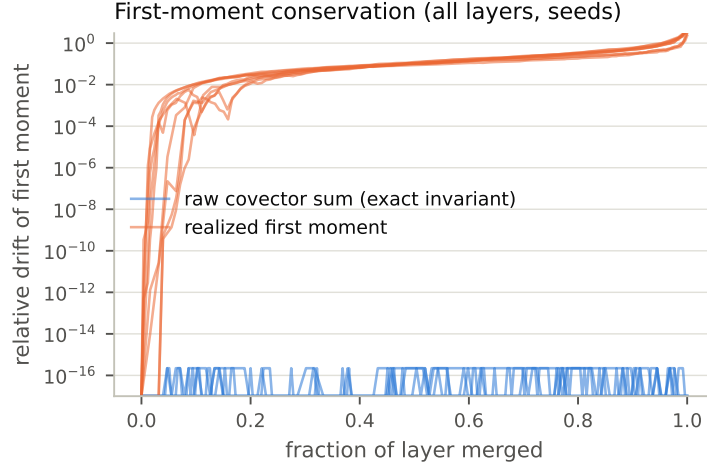


Figure 3: **Conservation laws in the data (Prop. 3.3, Rem. 3.7).** Relative drift of the layer’s total first moment along every merge trajectory (MNIST-MLP, all layers and seeds, log scale). The raw covector sum is conserved *exactly* — flat at the 10^{-16} machine-precision floor through the full trajectory, as Proposition 3.3 requires. The *realized* first moment (computed from the renormalized merged units) deviates at second order in cluster dispersion: negligible while genuine clusters are being consumed, growing to $O(1)$ only in the forced-merge regime — the overshoot factor $A_C / \|S_u\|$ of Remark 3.7 made visible.

Table 4: **E10–E13 (Phase D, CNNs):** the gap and its complete decomposition, capacity @−1pt (3-seed medians; fine ± 0.035 grid for the last three rows). Middle: removal-set overlap with OSSCAR at matched k (E11; chance in parentheses) — both methods drop largely the same filters even where capacities diverge, isolating the surgery as the cause.

arm	MNIST	Fashion	overlap @25% (chance 0.25)
merged units + kernel repair (E10)	0.174	0.326	0.50–0.75
our support + exact LS (E12)	0.482	0.580	”
+ swap refinement (E13)	0.549	0.616	”
OSSCAR, 128 imgs	0.549	0.616	1.00 (self)
MLP reference (E11)	overlap up to 0.94 at half the layer removed (chance 0.50)		